

Sri Lanka Institute of Information Technology



IT3010 –Network Design & Management 3rd Year, 2nd Semester Practical 1 - Lab Report

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B.Sc. (Hons) in Information Technology

Practical 1

✓ Virtualization Providers

- VMWare
- VirtualBox
- Hyper-V

Virtualization Provider	Virtual Interface Cards	Description
VMware Workstation	VMXNET3	High-performance virtual NICs designed for VMware VMs, providing better throughput and lower CPU usage.
Oracle VirtualBox	Intel PRO/1000 MT	A widely supported virtual NIC that emulates a real Intel Ethernet adapter, ensuring broad OS compatibility.
Microsoft Hyper-V	Legacy Network Adapter	A synthetic network adapter designed for optimized performance and integration with Windows-based virtual machines.

✓ Comparison of Virtual Interface Cards

Feature	VMXNET3 (VMware)	Intel PRO/1000 MT (VirtualBox)	Hyper-V Virtual Adapter
Performance	High-performance, low CPU usage	Moderate, depends on emulation	High-performance, optimized for Windows
OS Compatibility	Requires VMware Tools for full functionality	Broad OS support, works without guest additions	Best for Windows, Linux support requires drivers
Offloading Support	Yes (checksum, TCP segmentation, etc.)	Limited	Yes (RSS, VMQ, SR-IOV support)
Use Case	Best for production workloads in VMware	Best for compatibility and legacy systems	Best for Windows-based virtualization

✓ Practical Scenarios

- Scenario 1: VMware Workstation -(VMXNET3 Adapter)**
 A company wants to set up a high-speed virtualized web server on VMware Workstation. Since **VMXNET3** offers **high performance and lower CPU overhead**, it is used to maximize data transfer speeds between VMs and external clients.
- Scenario 2: VirtualBox – (Intel PRO/1000 MT Adapter)**
 A university sets up a VirtualBox environment for networking students. Since Intel PRO/1000 MT is compatible with most OS, it ensures all students can experiment with networking configurations without additional drivers.
- Scenario 3: Hyper-V – (Legacy Adapter)**
 A business needs to migrate an old Windows Server 2003 to a Hyper-V environment. Since **Legacy Network Adapter** supports older operating systems without extra drivers, it is used to ensure network connectivity during migration.

Lab sheet 1

1. Create a Directory called **NDM** in E drive of your PC.
2. Create another two sub-directories called **NDM-Server** and **NDM-Client** .
3. Download the CentOS and Fedora for the machine
4. for the VMs
 - a. Server – CentOS7
 - b. Client – Fedora 28
5. Create the VM for the server

New Virtual Machine Wizard

Name the Virtual Machine
What name would you like to use for this virtual machine?

Virtual machine name:
MLB-DC1-CentOs7

Location:
C:\Users\VENATH\Desktop\NDM\NDM-Server Browse...

The default location can be changed at Edit > Preferences.

< Back Next > Cancel

New Virtual Machine Wizard

×

Processor Configuration
Specify the number of processors for this virtual machine.

Processors

Number of processors: 1

Number of cores per processor: 1

Total processor cores: 1

Help

< Back

Next >

Cancel

New Virtual Machine Wizard

×

Specify Disk Capacity
How large do you want this disk to be?

Maximum disk size (GB): 20.0

Recommended size for CentOS 6 64-bit: 20 GB

☐ Allocate all disk space now.
Allocating the full capacity can enhance performance but requires all of the physical disk space to be available right now. If you do not allocate all the space now, the virtual disk starts small and grows as you add data to it.

☐ Store virtual disk as a single file
☒ Split virtual disk into multiple files
Splitting the disk makes it easier to move the virtual machine to another computer but may reduce performance with very large disks.

Help

< Back

Next >

Cancel

New Virtual Machine Wizard

×

Memory for the Virtual Machine

How much memory would you like to use for this virtual machine?

Specify the amount of memory allocated to this virtual machine. The memory size must be a multiple of 4 MB.

128 GB

64 GB

32 GB

16 GB

8 GB

4 GB

2 GB

1 GB

512 MB

256 MB

128 MB

64 MB

32 MB

16 MB

8 MB

4 MB

Memory for this virtual machine:

2048

MB

Maximum recommended memory:

13.3 GB

Recommended memory:

1 GB

Guest OS recommended minimum:

512 MB

Help

< Back

Next >

Cancel

ROOT PASSWORD

CENTOS 7 INSTALLATION

Done

us

Help

The root account is used for administering the system. Enter a password for the root user.

Root Password:

.....

Too short

Confirm:

.....

⚠ The passwords do not match.

Full name

svr1

User name

svr1

Tip: Keep your user name shorter than 32 characters and do not use spaces.

☐ Make this user administrator

☒ Require a password to use this account

Password

••••

Weak

Confirm password

••••

Advanced...

 The password you have provided is weak: The password contains the user name in some form. You will have to press Done twice to confirm it.

6. Create the VM for Client

New Virtual Machine Wizard

×

Name the Virtual Machine
What name would you like to use for this virtual machine?

Virtual machine name:

NDM-CLI-Fedora28

Location:

C:\Users\VENATH\Desktop\NDM\NDM-Client

Browse...

The default location can be changed at Edit > Preferences.

< Back

Next >

Cancel

New Virtual Machine Wizard

Processor Configuration

Specify the number of processors for this virtual machine.

Processors

Number of processors:

1

Number of cores per processor:

1

Total processor cores:

1

Help

< Back

Next >

Cancel

New Virtual Machine Wizard

Specify Disk Capacity

How large do you want this disk to be?

Maximum disk size (GB):

20.0

Recommended size for CentOS 6 64-bit: 20 GB

☐ Allocate all disk space now.

Allocating the full capacity can enhance performance but requires all of the physical disk space to be available right now. If you do not allocate all the space now, the virtual disk starts small and grows as you add data to it.

☐ Store virtual disk as a single file

☒ Split virtual disk into multiple files

Splitting the disk makes it easier to move the virtual machine to another computer but may reduce performance with very large disks.

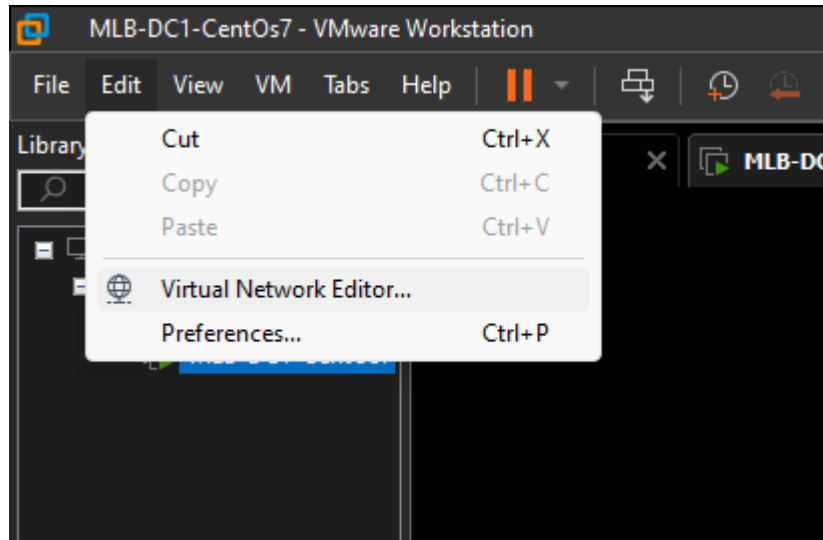
Help

< Back

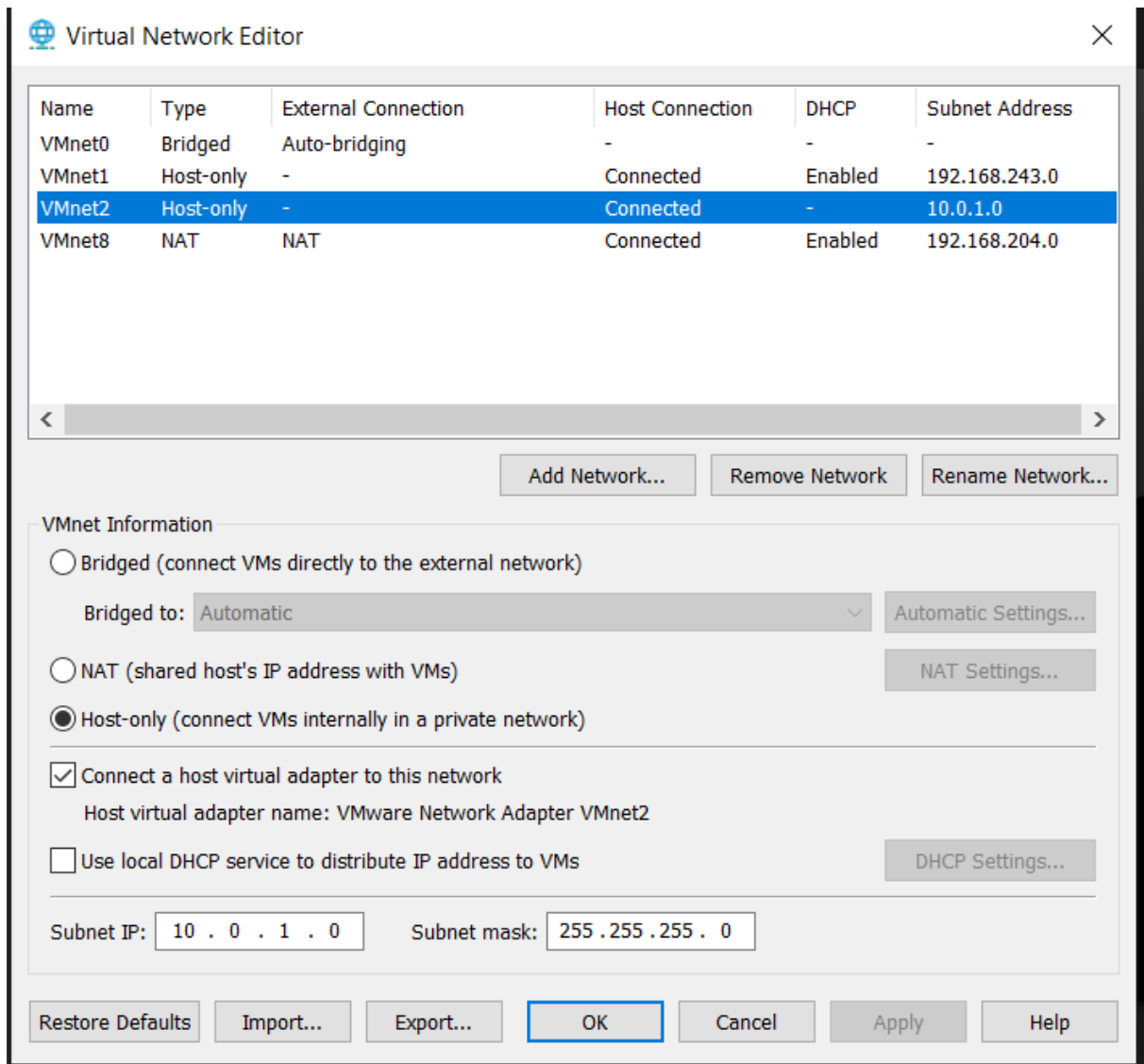
Next >

Cancel

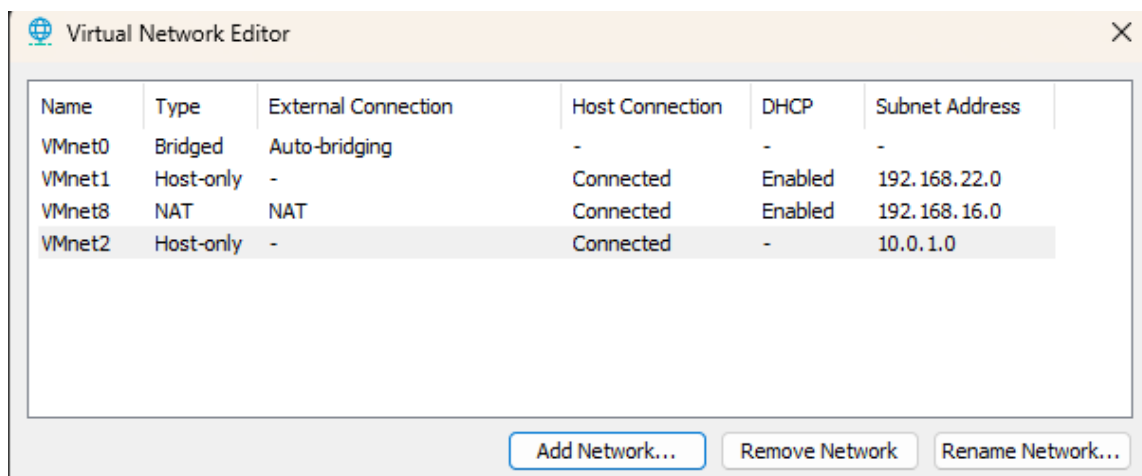
- ♦ after setting up, we need to configure the network
- ♦ for that we need to create the virtual switch
- ♦ then click to on the Edit in the nav bar



- then navigate to the Virtual Network Editor then it
- ♦ will open as follow



- Add a network for the VMnet2
- after that you need to configure the subnet as above and uncheck the “Use local DHCP service to distribute IP address to VMs”



- Then logging to the server to configure the ip address
- and check the ip address using following command on the terminal

`ip addr # return the all network interfaces`

- then check whether are there any interface assigned to the MAC Address
 - server mac - 00:0C:29:67:E8:D9
 - Client mac - 00:0C:29:F1:A3:80 - ens160

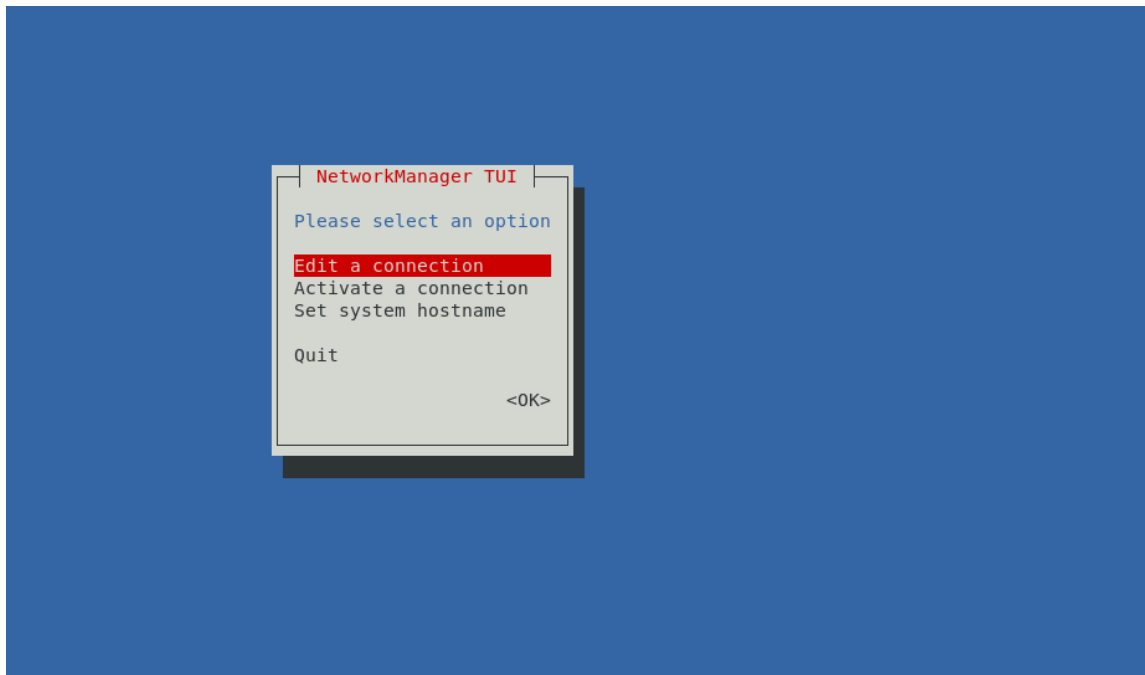
Network Configuration

Server side

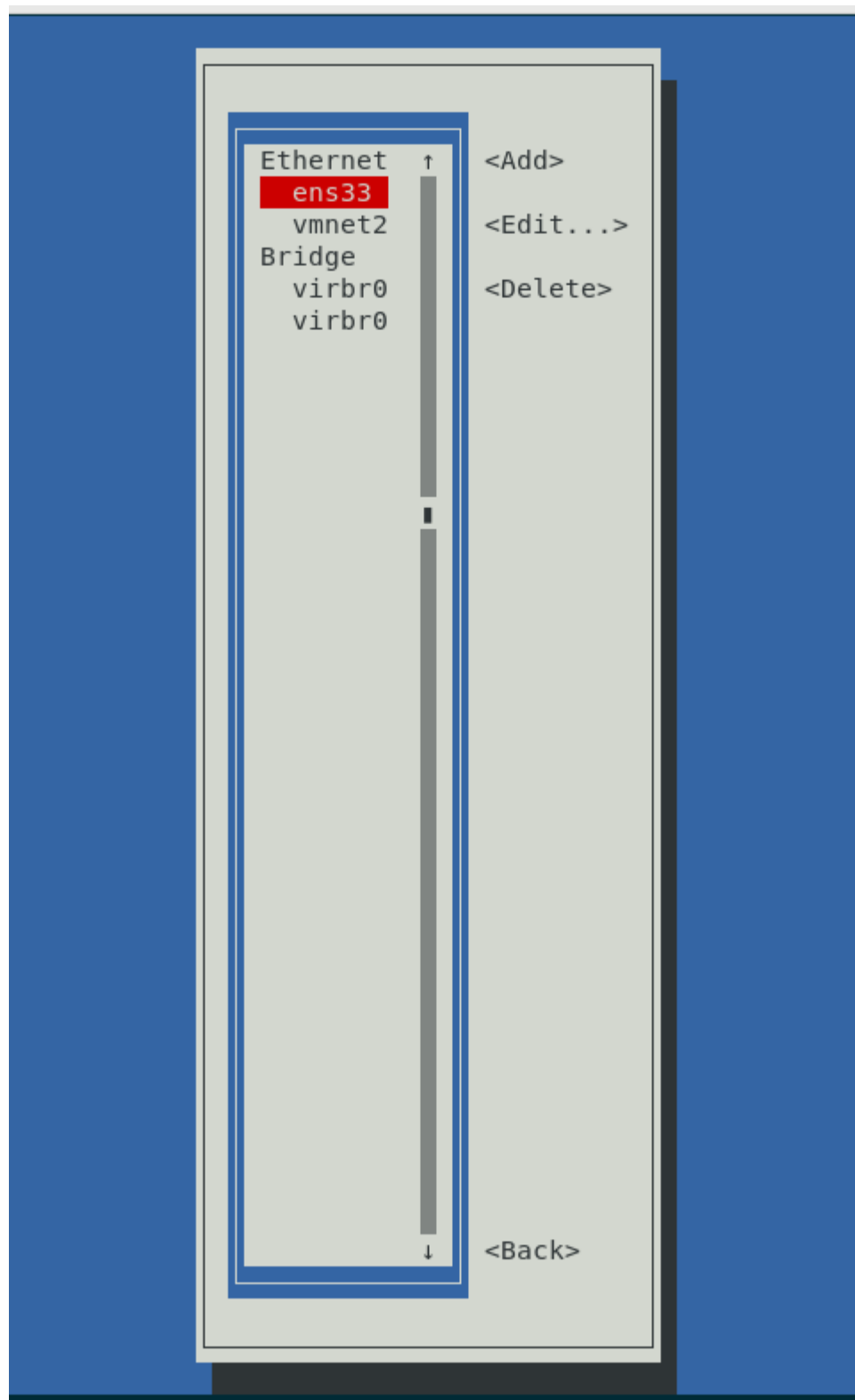
- open the terminal
- then get the interfaces that attached to the VM
 - server mac - 00:0C:29:67:E8:D9
- then check the interface availability of the VM
 - use command `ifconfig`

```
[root@localhost ~]# ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 00:0c:29:67:e8:d9 brd ff:ff:ff:ff:ff:ff
    inet 192.168.22.128/24 brd 192.168.22.255 scope global noprefixroute dynamic ens33
        valid_lft 1421sec preferred_lft 1421sec
    inet6 fe80::6a4b:53c0:7b24:6c47/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
3: ens34: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 00:0c:29:67:e8:e3 brd ff:ff:ff:ff:ff:ff
4: virbr0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc noqueue state DOWN group default qlen 1000
    link/ether 52:54:00:6d:d1:9f brd ff:ff:ff:ff:ff:ff
    inet 192.168.122.1/24 brd 192.168.122.255 scope global virbr0
        valid_lft forever preferred_lft forever
5: virbr0-nic: <BROADCAST,MULTICAST> mtu 1500 qdisc pfifo_fast master virbr0 state DOWN group default qlen 1000
    link/ether 52:54:00:6d:d1:9f brd ff:ff:ff:ff:ff:ff
[root@localhost ~]#
```

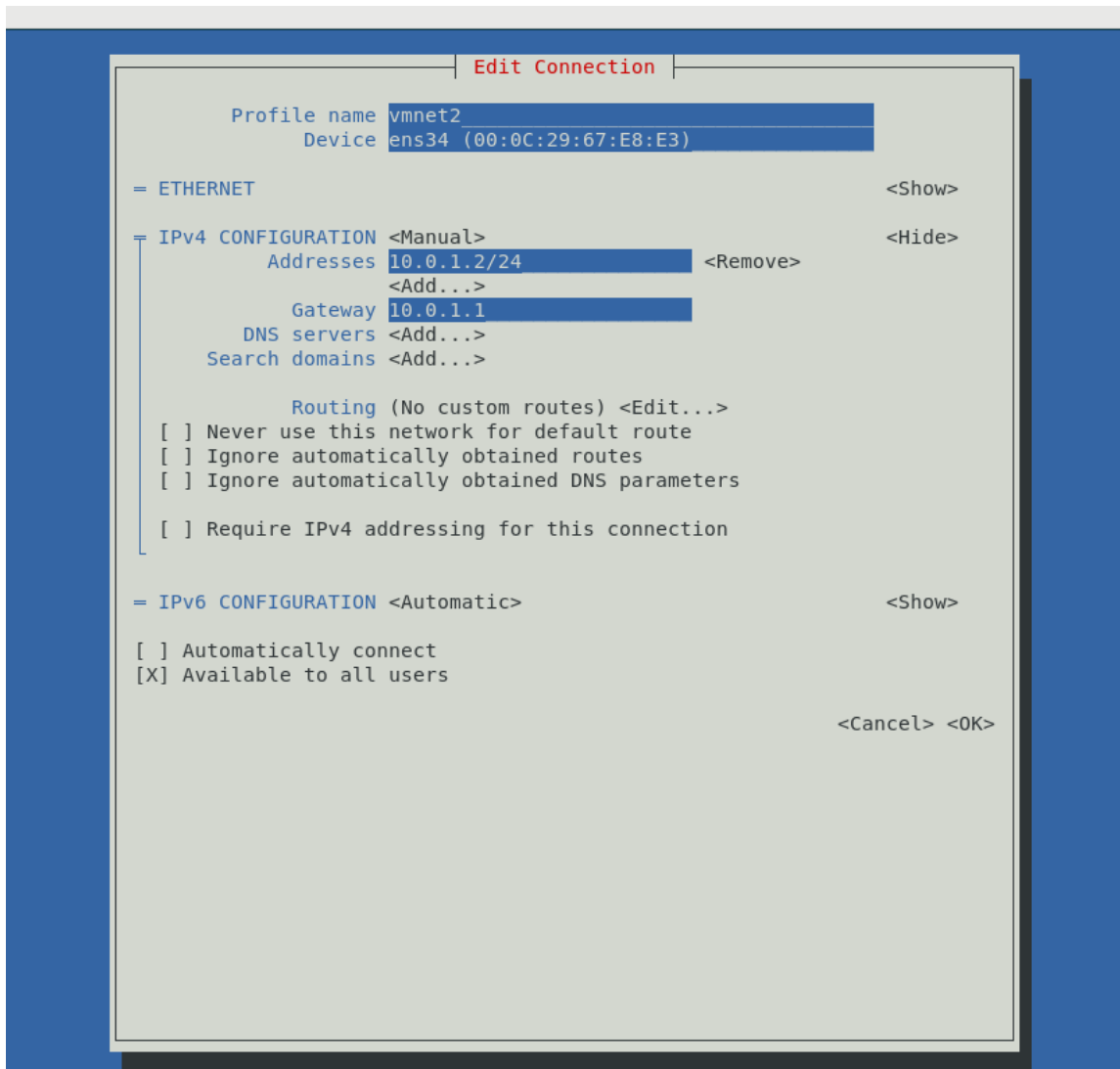
- then type `nmtui` to open the network interface configuration to edit the VM interfaces and the IP address



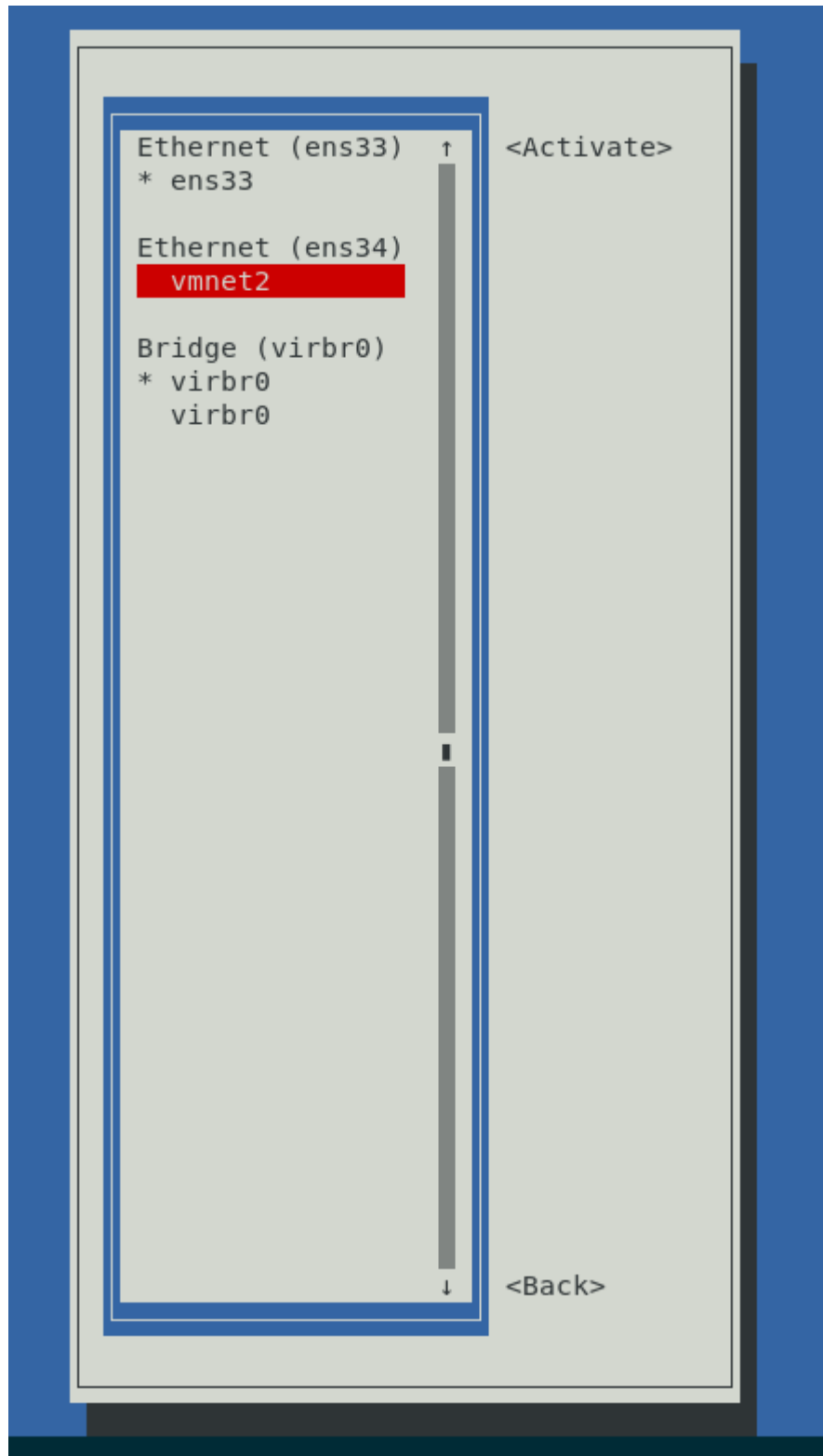
- ♦ then navigate to the Edit a Connection option



- ◊ Then edit the configurations as follow



- And then activate the vmnet2 interface to access the vm network

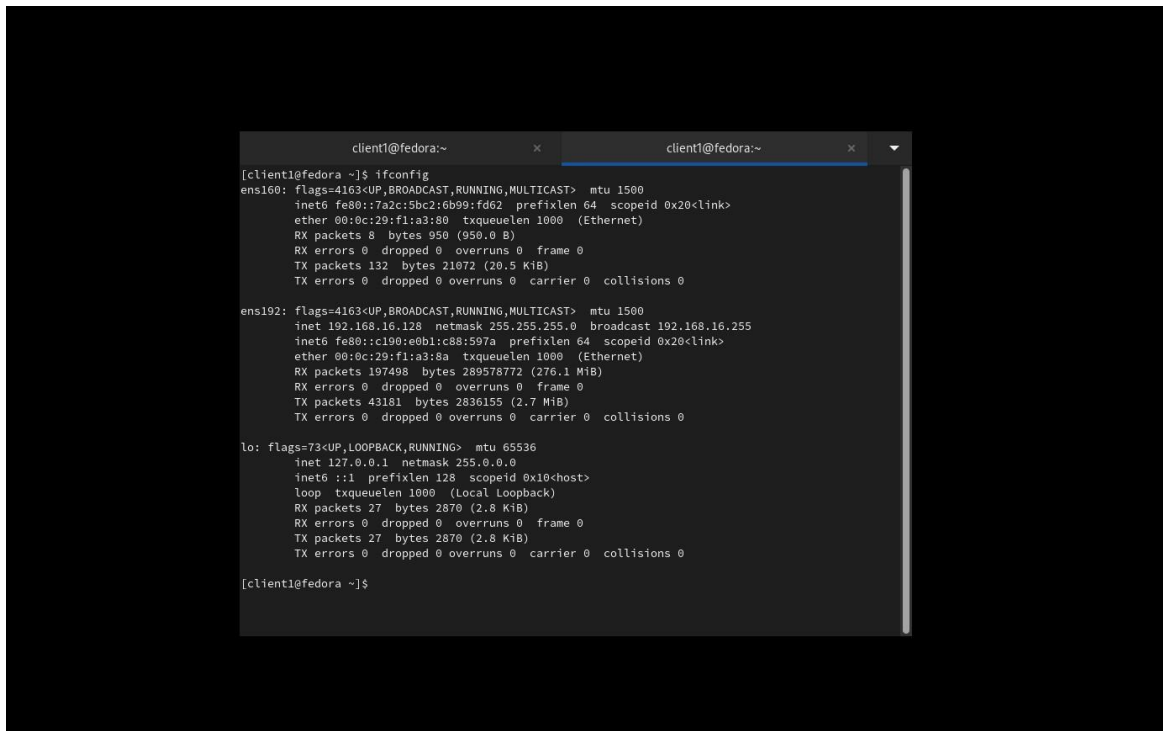


Client side - Fedora using terminal

- first spin up the fedora VM •

And open the terminal

- type `ifconfig` to get the network interfaces that we have merged to the VM



```
client1@fedora:~  
[client1@fedora ~]$ ifconfig  
ens160: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500  
    inet6 fe80::7a2c:5bc2:6b99:fd62 prefixlen 64 scopeid 0x20<link>  
    ether 00:0c:29:f1:a3:80 txqueuelen 1000 (Ethernet)  
    RX packets 0 bytes 950 (950.0 B)  
    RX errors 0 dropped 0 overruns 0 frame 0  
    TX packets 132 bytes 21072 (20.5 KiB)  
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0  
  
ens192: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500  
    inet 192.168.16.128 netmask 255.255.255.0 broadcast 192.168.16.255  
    inet6 fe80::c190:e0b1:c88:597a prefixlen 64 scopeid 0x20<link>  
    ether 00:0c:29:f1:a3:8a txqueuelen 1000 (Ethernet)  
    RX packets 197498 bytes 289578772 (276.1 MiB)  
    RX errors 0 dropped 0 overruns 0 frame 0  
    TX packets 43181 bytes 2836155 (2.7 MiB)  
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0  
  
lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536  
    inet 127.0.0.1 netmask 255.0.0.0  
    inet6 ::1 prefixlen 128 scopeid 0x10<host>  
    loop txqueuelen 1000 (Local Loopback)  
    RX packets 27 bytes 2870 (2.8 KiB)  
    RX errors 0 dropped 0 overruns 0 frame 0  
    TX packets 27 bytes 2870 (2.8 KiB)  
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0  
  
[client1@fedora ~]$
```

- Then note down the interface that is mapped with the MAC address of the Custom VMNAt2 -
 - Client mac - 00:0C:29:F1:A3:80
- then type open the Network Manger to configure the network addresses and gateway to the VM

```
nmtui
```

- If the Network manger is not available it return a command not found error then use following command to install the Networkmanager to the fedora system

```
sudo yum install NetworkManager-tui
```

```
#2) Think before you type.
#3) With great power comes great responsibility.

[sudo] password for client1:
Fedora 34 openh264 (From Cisco) - x86_64          690 B/s | 2.5 kB    00:03
Fedora Modular 34 - x86_64                      583 kB/s | 4.9 MB   00:08
Fedora Modular 34 - x86_64 - Updates             602 kB/s | 4.7 MB   00:08
Fedora 34 - x86_64 - Updates                     521 kB/s | 34 MB    01:07
Fedora 34 - x86_64                               1.6 MB/s | 74 MB    00:45
Dependencies resolved.
=====
Package                        Architecture Version      Repository    Size
=====
Installing:
NetworkManager-tui            x86_64      1:1.30.6-1.fc34 updates      215 k
Upgrading:
NetworkManager                x86_64      1:1.30.6-1.fc34 updates      2.3 M
NetworkManager-adsl           x86_64      1:1.30.6-1.fc34 updates       26 k
NetworkManager-bluetooth      x86_64      1:1.30.6-1.fc34 updates       52 k
NetworkManager-libnm          x86_64      1:1.30.6-1.fc34 updates      1.6 M
NetworkManager-ppp            x86_64      1:1.30.6-1.fc34 updates       35 k
NetworkManager-team           x86_64      1:1.30.6-1.fc34 updates       30 k
NetworkManager-wifi           x86_64      1:1.30.6-1.fc34 updates      106 k
NetworkManager-wwan           x86_64      1:1.30.6-1.fc34 updates       58 k
glib2                         x86_64      2.68.4-1.fc34 updates      2.6 M
```

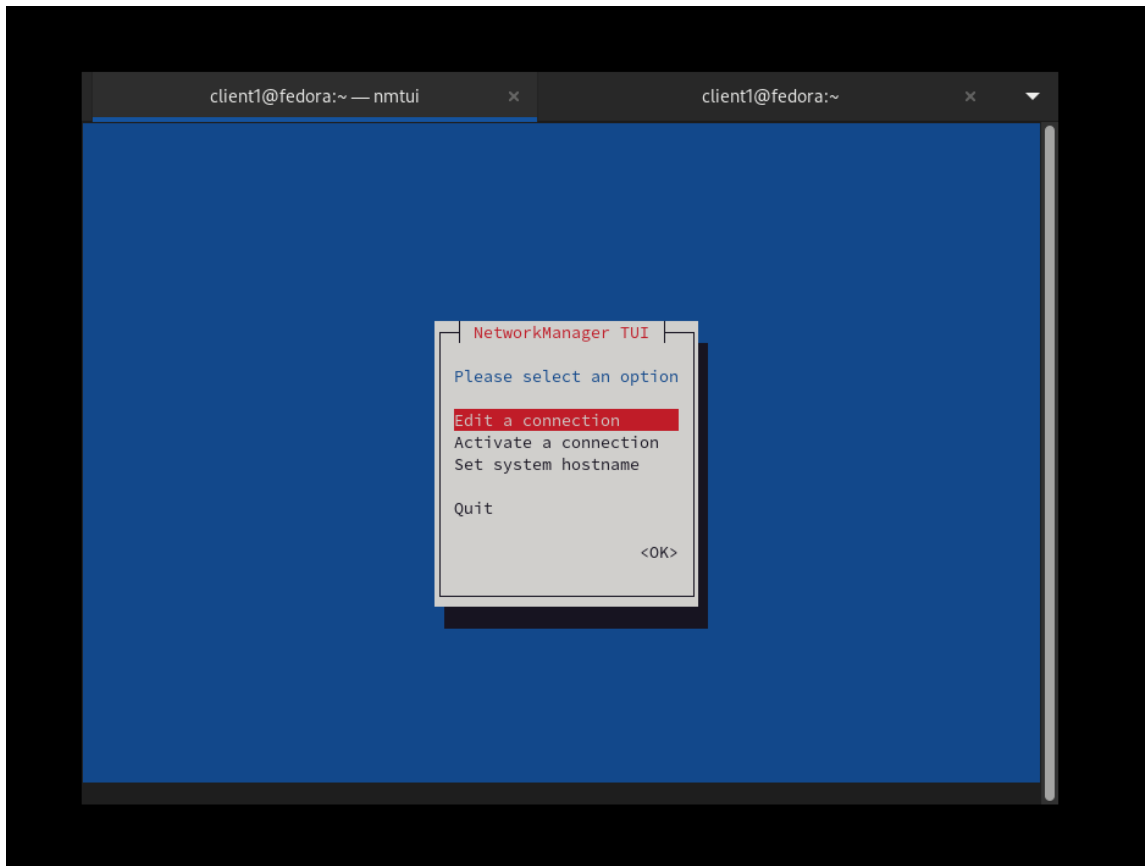
```
client1@fedora:~ x client1@fedora:~ x
Verifying : NetworkManager-1:1.30.6-1.fc34.x86_64 4/21
Verifying : NetworkManager-1:1.30.2-1.fc34.x86_64 5/21
Verifying : NetworkManager-adsl-1:1.30.6-1.fc34.x86_64 6/21
Verifying : NetworkManager-adsl-1:1.30.2-1.fc34.x86_64 7/21
Verifying : NetworkManager-bluetooth-1:1.30.6-1.fc34.x86_64 8/21
Verifying : NetworkManager-bluetooth-1:1.30.2-1.fc34.x86_64 9/21
Verifying : NetworkManager-libnm-1:1.30.6-1.fc34.x86_64 10/21
Verifying : NetworkManager-libnm-1:1.30.2-1.fc34.x86_64 11/21
Verifying : NetworkManager-ppp-1:1.30.6-1.fc34.x86_64 12/21
Verifying : NetworkManager-ppp-1:1.30.2-1.fc34.x86_64 13/21
Verifying : NetworkManager-team-1:1.30.6-1.fc34.x86_64 14/21
Verifying : NetworkManager-team-1:1.30.2-1.fc34.x86_64 15/21
Verifying : NetworkManager-wifi-1:1.30.6-1.fc34.x86_64 16/21
Verifying : NetworkManager-wifi-1:1.30.2-1.fc34.x86_64 17/21
Verifying : NetworkManager-wwan-1:1.30.6-1.fc34.x86_64 18/21
Verifying : NetworkManager-wwan-1:1.30.2-1.fc34.x86_64 19/21
Verifying : glib2-2.68.4-1.fc34.x86_64 20/21
Verifying : glib2-2.68.1-1.fc34.x86_64 21/21

Upgraded:
NetworkManager-1:1.30.6-1.fc34.x86_64 NetworkManager-adsl-1:1.30.6-1.fc34.x86_64
NetworkManager-bluetooth-1:1.30.6-1.fc34.x86_64 NetworkManager-libnm-1:1.30.6-1.fc34.x86_64
NetworkManager-ppp-1:1.30.6-1.fc34.x86_64 NetworkManager-team-1:1.30.6-1.fc34.x86_64
NetworkManager-wifi-1:1.30.6-1.fc34.x86_64 NetworkManager-wwan-1:1.30.6-1.fc34.x86_64
glib2-2.68.4-1.fc34.x86_64

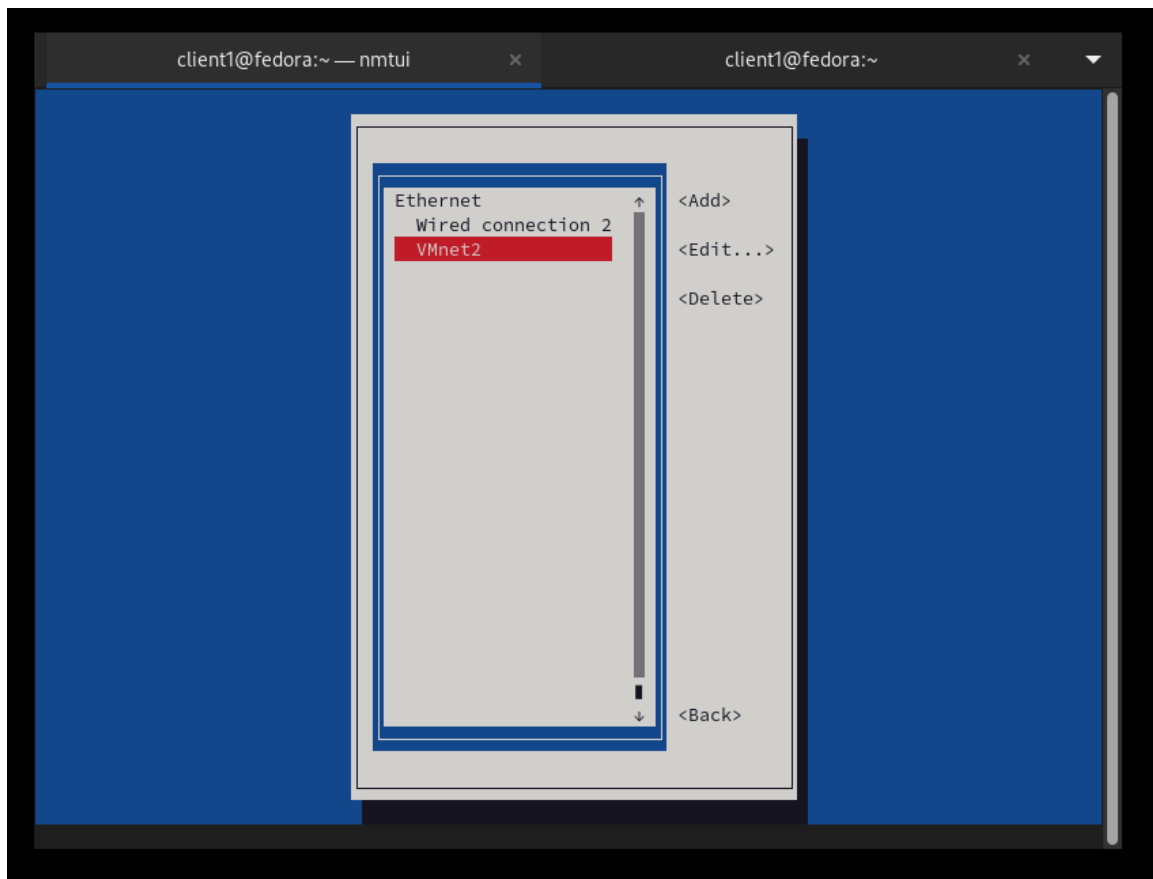
Installed:
NetworkManager-tui-1:1.30.6-1.fc34.x86_64 newt-0.52.21-9.fc34.x86_64 slang-2.3.2-9.fc34.x86_64

Complete!
[client1@fedora ~]$
```

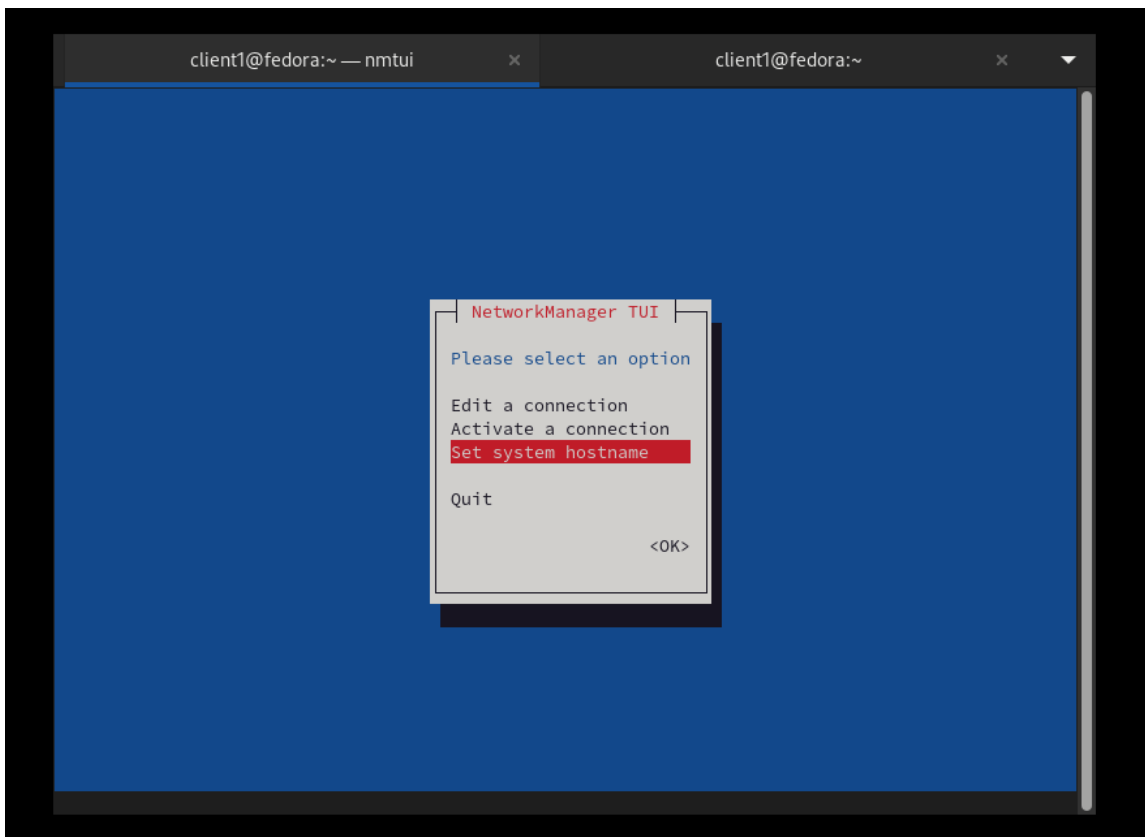
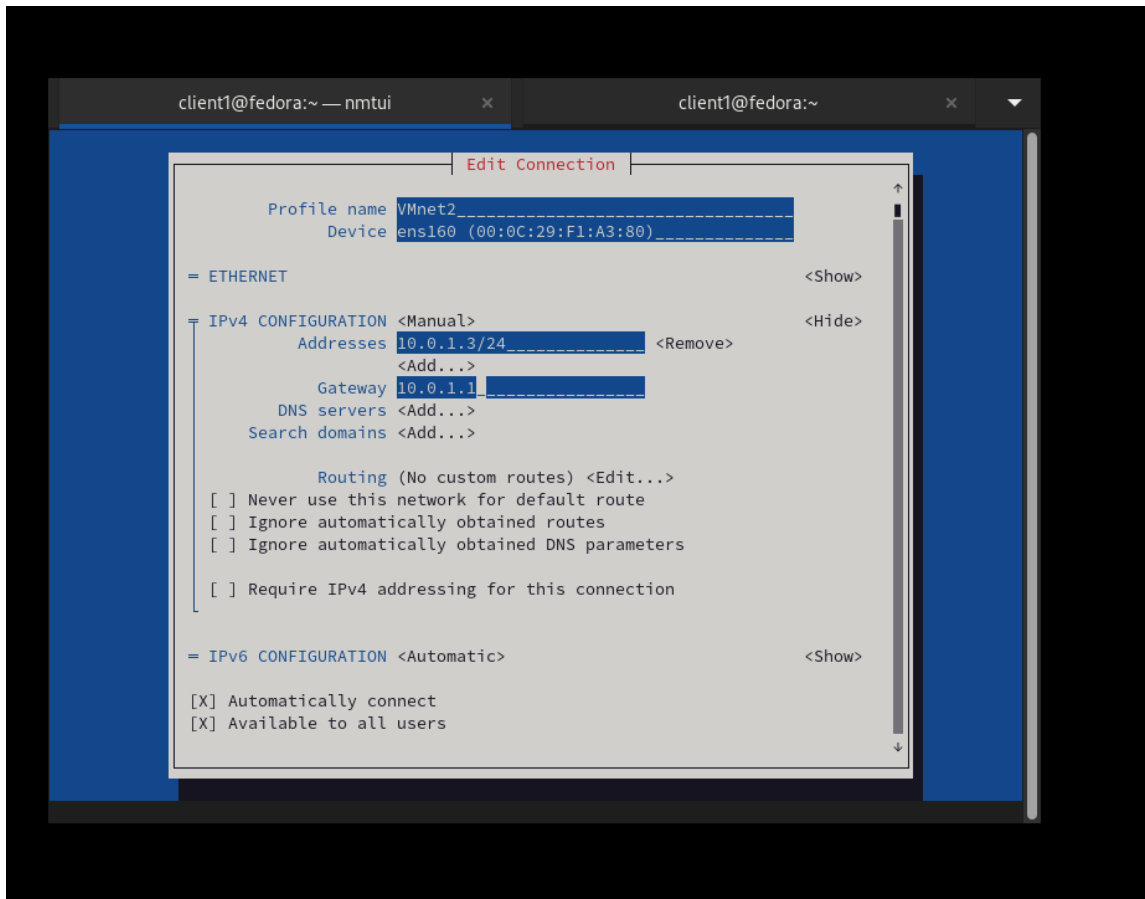
- it will install the `nmtui` to the system



- ♦ navigate to the edit connection ◦
Select the VMnet2



- And edit the ipaddress and the gateway and navigate back to the main window



- Go to activate connection and
- activate the VMnet2

